

Joining Forces or Going It Alone? On the Interplay among External Collaboration Partner Types, Interfirm Governance Modes, and Internal R&D*

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This paper adds to the emerging literature stream advocating a contingency view on open innovation. Drawing on the relational view of the firm, this study sheds light on the complex interplay among collaboration partner types (market- and science-focused innovation partners), governance modes (informal, self-enforcing and formal, contractual collaboration governance), and internal research and development (R&D). More specifically, it is proposed that the use of governance modes tailored to both the characteristics of each innovation partner type and the specific innovation objectives pursued by the focal firm (incremental and radical new product development) can increase the payoff from innovation collaboration. Moreover, appropriate collaboration governance is expected to reduce the focal firm's vulnerability to possible negative side effects often assumed to be associated with the simultaneous pursuit of external collaboration and internal R&D, among which most notably the not-invented-here (NIH) syndrome. Cross-industry evidence from 2502 German firms underlines the critical role of collaboration governance—a contingency factor that is at the heart of the relational view, yet has remained surprisingly absent from the open innovation debate so far.

Introduction

Firms competing in today's global market face the challenge of developing new products and services not only more quickly (time to market), but also more economically (cost to market) than in the past, while striving for greater novelty (new to market) and customer appeal (fit to market) (Reichwald and Piller, 2009; Salge, Bohné, Farchi, and Piening, 2012). To master this difficult task, scholars and practitioners alike are increasingly advocating an open approach to new product development (NPD). This requires firms to systematically perform “knowledge exploration, retention, and exploitation inside and outside an organization's boundaries throughout the innovation process” (Lichtenthaler, 2011, p. 77). The open innovation paradigm advises firms to reach beyond their own boundaries to join forces with customers, suppliers, competitors, universities, and other research organizations (Chesbrough, 2003; Love, Roper, and Vahter, 2013; Rosenkopf and

Almeida, 2003). By partnering with a diverse set of external actors, firms are assumed to increase the likelihood of gaining access to valuable new knowledge and complementary assets, thereby boosting their innovative performance (e.g., Dahlander and Gann, 2010; Lichtenthaler, 2011).

That said, recent evidence casts doubt on the prevalent assumption that openness is a panacea for achieving NPD success. Increasing the number of collaboration partners is not only associated with growing knowledge identification, assimilation, and utilization costs (Belderbos, Faems, Leten, and Van Looy, 2010; Laursen and Salter, 2006; Salge, Farchi, Barrett, and Dopson, 2013), but also entails possible appropriation concerns (Alexy, George, and Salter, 2013; Almirall and Casadesus-Masanell, 2010). These arguments gave rise to the recent emergence of a small, but rapidly growing literature stream advocating a contingency perspective on open innovation. Research in this stream seeks to unearth the subtle influences of firms' specific innovation objectives (Leiponen and Helfat, 2010), innovation partner types (Sofka and Grimpe, 2010), innovation management activities (Salge et al., 2012), internal structures and constraints (Arora, Belenzon, and Rios, 2014; Garriga, von Krogh, and Spaeth, 2013), NPD project characteristics (Salge et al., 2013), and external boundary conditions such as environmental dynamism or appropriability regimes in shaping firms' return from open innovation (e.g., Lichtenthaler, 2011). Despite the substantial progress made within this

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emerging literature segment in recent years, research on the contingencies of open innovation is still in its infancy, rendering our understanding of how, why, and when openness is beneficial or detrimental for NPD incomplete (Hsieh and Tidd, 2012; Huizingh, 2011).

This paper seeks to contribute to this debate in at least three meaningful ways. First, it helps strengthen the theoretical foundations underpinning current thinking on the boundary conditions of open innovation. To do so, it merges conceptual ideas from the open innovation litera-

ture with the relational view of the firm that specifies the conditions under which firms are more or less likely to realize the potential benefits of interfirm collaboration (Dyer and Singh, 1998; Lavie, 2006). Adopting such an integrative theoretical lens helps illuminate the complex interplay among *collaboration partner types* (market- and science-focused innovation partners), *governance modes* (informal, self-enforcing and formal, contractual collaboration governance), and *internal research and development (R&D)*. Second and more specifically, this study extends the emerging contingency stream in the open innovation literature by illuminating the critical role of collaboration governance—a contingency factor that is at the heart of the relational view, yet has remained surprisingly absent from the open innovation debate so far. In doing so, it uncovers a twofold moderating effect of collaboration governance on the payoff from innovation collaboration. In particular, formal governance mechanisms hold potential to enhance a focal firm's ability to capture value from collaboration, while informal mechanisms help protect it from the negative side effects frequently accompanying the simultaneous pursuit of external collaboration and internal R&D. These insights in turn enhance not only open innovation research, but also relational view of the firm in that they inform the ongoing debate on the relative effectiveness of formal, contract-based and informal, self-enforcing governance (Vlaar, Van den Bosch, and Volberda, 2007). Third and building on the previous contribution, this study sheds new light on the relationship between firms' internal R&D and external knowledge sourcing, which continues to puzzle scholars in the field of product innovation management (Hagedoorn and Wang, 2012; Vega-Jurado, Gutiérrez-Gracia, and Fernández-de-Lucio, 2009). While the importance of combining knowledge from internal and external sources in the development of innovative products and services has been recognized for many years, both theory and evidence remain ambiguous as to whether external innovation collaboration and internal R&D activities act as complements (Cassiman and Veugelers, 2006) or substitutes (e.g., Laursen and Salter, 2006). By introducing partner type and governance mode as two new contingency variables, this study contributes to clarifying the conditions under which complementarities or substitution effects between external collaboration and internal R&D are likely to arise. These insights are not only of academic interest, but also provide actionable knowledge to product innovation managers seeking to structure and govern their innovation collaboration portfolio such that it contributes best to achieving their specific innovation objectives.

BIOGRAPHICAL SKETCHES

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Theory and Hypotheses

The resource-based view (RBV) of the firm suggests that the firm's competitiveness and innovativeness depends on its internal, firm-specific resources and capabilities (Penrose, 1959; Wernerfelt, 1984). The resources owned or controlled by the firm specifically constitute a source of competitive advantage if they fulfill the criteria of being valuable, rare, inimitable, and nonsubstitutable (Barney, 1991). Complementing the inward-looking perspective of the RBV, the relational view proposed by Dyer and Singh (1998) proceeds from the assumption that strategic resources can also span firm boundaries. This theoretical perspective highlights the importance of interorganizational collaboration as a source of competitive advantage and relational rents, that is, "a supernormal profit jointly generated in an exchange relationship that cannot be generated by either firm in isolation" (Dyer and Singh, 1998, p. 662). The implication is that by combining their idiosyncratic resources in unique ways, firms engaging in collaborative arrangements will enjoy greater innovation success than those that innovate inside their boundaries (Duschek, 2004; Dyer and Singh, 1998; Lavie, 2006). As Powell, Koput, and Smith-Doerr (1996, p. 116) put it: "the locus of innovation will be found in networks of learning, rather than in individual firms."

The relational view provides a valuable theoretical foundation for the study of open innovation, especially because prior work on this issue has been criticized for its lack of theoretical grounding (Lichtenthaler, 2011). In contrast to the traditional RBV whose proprietary assumption (i.e., a firm should protect its resources from imitation) appears to somewhat contradict the idea of opening up the innovation process, the relational view specifically addresses the sources of collaborative rent generation (Duschek, 2004; Ketchen, Ireland, and Snow, 2007; Lavie, 2006). According to this theoretical framework, firms will be more likely to realize the potential benefits of their collaborative innovation activities when they possess (1) partner-specific *absorptive capacity* and (2) complementary *resource endowments* and (3) when they employ effective *governance mechanisms* (e.g., Dyer and Singh, 1998; Lavie, 2006).

First, extending Cohen and Levinthal's (1990) widely recognized absorptive capacity concept, according to which the focal firm's ability to harness external knowledge is a function of its prior related knowledge generated through internal R&D, Dyer and Singh (1998) introduce the idea of partner-specific absorptive capacity. They argue that the focal firm's ability to generate value from its innovation collaborations is not only dependent

upon its general absorptive capacity, as shown by previous studies in the context of open innovation (e.g., Salge et al., 2012), but may be particularly affected by its ability to absorb valuable knowledge from a specific collaboration partner. Among others, this capability is a function of the degree of overlap in the knowledge bases of collaboration partners (Dyer and Singh, 1998; see also Ritala and Hurmelinna-Laukkanen, 2013). Second, complementary resource endowments are another important dimension of collaboration success (Duschek, 2004; Dyer and Singh, 1998), which has been extensively discussed in the innovation literature (e.g., Teece, 1986). Product innovation often requires that the focal firm's resources (e.g., technologies) are used in combination with a collaboration partner's idiosyncratic resources (e.g., local market knowledge, distribution channels). By selecting collaboration partners with specific resources, the focal firm can leverage the value of its own resources. This effect is exemplified by start-up firms that enhance their reputation through collaborating with prestigious partners (Lavie, 2006; Lin, Yang, and Arya, 2009). Finally, proponents of the relational view argue that the selection of collaboration partners also influences to what extent there is an alignment of incentives that encourages the partners to be transparent, exchange knowledge, and not act opportunistically (Duschek, 2004; Dyer and Singh, 1998). To ensure interfirm knowledge transfer, motivate collaboration partners to invest in relation-specific assets, and protect against the hazards of opportunism, firms can employ various governance modes, or what is also referred to as safeguards (e.g., Dyer, 1997). In essence, the literature distinguishes between formal governance mechanisms such as contracts that specify the rights as well as obligations of collaboration partners and rely on third-party enforcement (e.g., state authorities) and informal, self-enforcing mechanisms such as trust (Carson, Madhok, and Wu, 2006; Dyer and Singh, 1998; Poppo and Zenger, 2002). Although research suggests that informal agreements based on trust are generally more effective and involve lower transaction costs than formal safeguards, the choice of optimal governance modes nevertheless appears to be highly context dependent (Krishnan, Martin, and Noorderhaven, 2006; Poppo and Zenger, 2002).

Open Innovation and Relational Contingencies

The relational view aids in identifying theoretically meaningful contingencies that affect the effectiveness of the focal firm's open innovation activities. By merging the arguments advanced above with previous research on

open innovation, the *partner type* (market- versus science-focused innovation partners) and *governance mode* (informal versus formal) of innovation collaborations can be derived as salient factors in this respect.

The first dimension recognizes the specific characteristics of different types of collaboration partners (e.g., customers, suppliers, universities, and research institutes) in terms of their value-creating potential and value-appropriation hazards (e.g., Bercovitz and Feldman, 2007; Faems, Van Looy, and Debackere, 2005; Ritala and Hurmelinna-Laukkanen, 2013). In particular, following the relational view, these partners may not only vary in their incentives to act opportunistically, but also in their contribution to the focal firm's innovation processes given differences in the complementarity and transferability of their idiosyncratic resources (Dyer and Singh, 1998; Lavie, 2006). The literature suggests that collaboration partners can be broadly categorized into *market- and science-focused* innovation partners, depending on the kind of knowledge (market versus technological knowledge) they bring into the partnership (e.g., Bogers, Afuah, and Bastian, 2010; Spanjol, Qualls, and Rosa, 2011; Wiklund and Shepherd, 2003). On the one hand, market-focused innovation partners such as suppliers and customers (business customers or consumers) are widely expected to contribute to NPD success by providing the focal firm with a better understanding of potential applications of its technologies, market opportunities, and how to serve markets (e.g., marketing strategies) (Shane, 2000; Wiklund and Shepherd, 2003). Interacting with customers, for example, can enhance the market fit of new products or services through gaining an understanding of customers' needs and preferences ("need information") (Bonner and Walker, 2004; von Hippel and Katz, 2002). On the other hand, collaborations with science-focused innovation partners such as universities and research institutes are particularly valuable for assessing unique technological knowledge ("solution information") that is distinct from the focal firm's knowledge base. As these institutions typically not only engage in applied but also basic research, science-based collaborations hold promise for generating technological breakthroughs that offer long-term commercial potential (e.g., Bercovitz and Feldman, 2007; Faems et al., 2005; Grimpe and Kaiser, 2010).

Second, the issue of governance is at the heart of the relational view as the choice of a particular governance mode is likely to influence not only the transaction costs of collaborative arrangements, but also the behavior of collaboration partners including their willingness to share knowledge or their inclination to opportunistic behavior

(Duschek, 2004; Dyer, 1997). Accordingly, an effective governance mode is one that minimizes transaction costs while maximizing value-creation incentives (Dyer and Singh, 1998). Drawing on transaction costs economics (e.g., Williamson, 1985), two main approaches to safeguard against opportunistic behaviors of collaboration partners such as shirking, failing to fulfill obligations, and withholding valuable information are discussed, namely, a relation-based approach that emphasizes trust and the contract-based approach that emphasizes control (Carson et al., 2006; Lui and Ngo, 2004). Informal innovation collaboration reflects the former approach. Instead of using complex contracts that specify the roles and responsibilities to be performed as well as benefits and obligations of the focal firm and its collaboration partner, informal collaboration relies on self-enforcing mechanisms, most notably trust and reciprocity (Dyer, 1997; Poppo and Zenger, 2002; Vlaar et al., 2007). As mentioned above, under most conditions, this form of governance is viewed as more effective and less costly than formal collaboration based on contracts because of, for example, lower contracting and monitoring costs (Dyer and Singh, 1998). Yet, departing from Dyer and Singh's assumptions, other authors question whether informal innovation collaboration is universally superior to formal arrangements. Rather than undermining trust between collaboration partners, well-specified contracts may reduce uncertainty and risk, promote the establishment of long-lasting and cooperative relationships, and enhance the coordination of activities (Poppo and Zenger, 2002; Vlaar et al., 2007).

To conclude, the above discussion underscores the necessity to adopt a contingency perspective on open innovation. As a firm's decisions concerning its innovation partner types, governance modes, and innovation objectives appear to be mutually dependent, product innovation managers seeking to improve innovation success are well advised to carefully align these dimensions when configuring their firm's collaboration activities. Below, this assumption is elaborated in more detail.

Partner Type–Governance Mode Configurations and NPD Success

The need to align the governance modes to the specific collaborative arrangement at hand is a key tenet of the relational view adopted from Williamson's transaction cost economics (e.g., Lui and Ngo, 2004; Poppo and Zenger, 2002). In this regard, Dyer and Singh (1998) argue that the ability of exchange partners to match governance structures with exchange attributes is essential

for realizing relational rents. Relevant exchange attributes that determine the exchange hazards, and hence the choice of optimal governance modes, are the partner, level of uncertainty (e.g., environmental change, lack of information), and characteristics of the exchange relationship (e.g., frequency of interaction) (Carson et al., 2006; Lui and Ngo, 2004; Van de Vrande, Lemmens, and Vanhaverbeke, 2006). That said, given specific attributes, market- and science-based collaborations have different implications for the design of governance modes. For three main theoretical reasons, firms are likely (and might be well advised) to employ formal contractual arrangements for market-based collaborations and informal governance for science-based collaborations.

First, it is expected that the higher the *risk of opportunism*, the more formal the contractual arrangement of an innovation partnership to minimize the costs and performance losses arising from such hazards (e.g., Hurmelinna-Laukkanen, Olander, Blomqvist, and Panfilii, 2012; Lui and Ngo, 2004; Poppo and Zenger, 2002). In this respect, research suggests that market-based partners will be more inclined to misuse collaborative relationships for accessing and exploiting the focal firm's proprietary knowledge for their own advantage (Emden, Calantone, and Droge, 2006; Ritala and Hurmelinna-Laukkanen, 2013). Compared with business customers that might easily become future competitors of the focal firm, universities and other research institutions have fewer incentives to act opportunistically and also typically lack complementary assets required to commercialize intellectual property generated as an outcome of joint research (Bercovitz and Feldman, 2007; Martinez-Noya, Garcia-Canal, and Guillen, 2013). Moreover, the greater similarity between business firms in their structures, processes, and cultures as well as the greater overlap of knowledge compared with research institutions may not only facilitate voluntary interfirm knowledge transfer, as suggested by Dyer and Singh's (1998) theory, but likewise involuntary knowledge spillover (Emden et al., 2006). Contracts that specify procedures for monitoring and penalties for noncompliance might discourage market partners from using their opportunities to harm the focal firm (Lui and Ngo, 2004; Poppo and Zenger, 2002).

Second, market- and science-based collaboration are associated with different levels of *uncertainty*. While collaborations with customers are mainly directed at improving existing products incrementally (Faems et al., 2005), firms collaborate with universities and other research institutes to obtain fundamentally new knowledge and technologies stemming from basic research

(e.g., Grimpe and Kaiser, 2010). The latter innovation processes are characterized by a greater degree of ambiguity, as there is substantial uncertainty about the outcomes of the collaboration, lack of clear information about the technology to be developed, and uncertainty of available courses of action at the beginning of the partnership (Carson et al., 2006). Formal contractual regimes, which tend to be rather inflexible and not well suited for conditions of high ambiguity and environmental change (e.g., the costs for writing and enforcing contracts increase with uncertainty), appear to be the less desirable option for science-focused than for market-focused collaborations (see Carson et al., 2006; Leiblein and Miller, 2003). Indeed, evidence indicates that firms tend to rely on relational, self-enforcing governance modes when uncertainty is high (e.g., Crocker and Masten, 1996).

Finally, as collaborations with customers are typically characterized by a higher degree of *asset specificity* than those with research institutions that require low levels of transaction-specific investments, the former type is more likely to be governed by formal contracts (Carson et al., 2006; Poppo and Zenger, 2002). For example, when the focal firm collaborates with a client to develop customized products or services for this firm, the assets invested in this partnership might not be utilizable for other purposes (Jones, Hesterly, and Borgatti, 1997). Such asset specificity increases the costs of partner replacement, risk of opportunistic behavior, and demands for coordination (Williamson, 1985). As shown empirically, in line with transaction cost theorizing, firms respond with greater levels of contractual complexity to these issues (Poppo and Zenger, 2002).

In light of the arguments advanced thus far, it appears in fact plausible that formal governance is most effective for market-focused innovation collaboration, whereas informal, self-enforcing governance is most effective, both in terms of incremental and radical NPD success, for science-focused innovation collaboration. This conclusion challenges the widely held assumption that self-enforcing mechanisms are generally more effective than third-party enforcement (Dyer and Singh, 1998). That said, it is consistent with Dyer and Singh's (1998) recognition of the liabilities of informal governance and initial evidence revealing that both forms of governance can be effective under certain conditions (e.g., Carson et al., 2006; Poppo and Zenger, 2002). Thus:

H1: Formally governed, market-focused collaborations will be associated more strongly with firms' incremental and radical NPD success than their informally governed counterparts.

H2: Informally governed, science-focused collaborations will be associated more strongly with firms' incremental and radical NPD success than their formally governed counterparts.

Open Innovation, Internal R&D, and Collaboration Governance

Rather than solely relying on knowledge from external sources, firms still need to engage in their own internal R&D activities to successfully develop new products and services (e.g., Cassiman and Veugelers, 2006; Dahlander and Gann, 2010; Lokshin, Belderbos, and Carree, 2008). From a resource-based perspective, combining knowledge generated through internal R&D with knowledge from external sources in unique ways is indeed a particularly likely source of competitive advantage, as these knowledge combinations are assumed to be high in value and inimitability (Grimpe and Kaiser, 2010). However, the literature on this important issue is riddled with countervailing theoretical arguments and mixed empirical evidence. On the one hand, drawing on absorptive capacity theory, research developed a compelling case for complementarity between these innovation activities, as the firm's ability to absorb knowledge from external sources is assumed to be a function of its prior related knowledge stemming predominantly from internal R&D (Cohen and Levinthal, 1990). Moreover, internal R&D capabilities have been described as a prerequisite to attract competent collaboration partners (Dahlander and Gann, 2010). On the other hand, the assumption of substitution effects between internal R&D and external knowledge sourcing is primarily inspired by work on the not-invented-here (NIH) syndrome (Antons and Piller, forthcoming; Katz and Allen, 1982). The NIH syndrome refers to the presence of a profound, general bias against ideas generated outside the focal firm. This bias might be particularly prevalent in organizations with strong internal R&D capabilities. This negative attitude results from the deeply held belief of innovation managers and expert staff that knowledge and ideas generated internally are necessarily superior to external knowledge. External collaboration may question the legitimacy of the R&D department, leading to resistance of organizational members seeking to protect their interests (Katz and Allen, 1982; Lichtenthaler and Ernst, 2006). By providing support for the existence of both complementarity (Cassiman and Veugelers, 2006; Grimpe and Kaiser, 2010; Lokshin et al., 2008) and substitution (Hess and Rothaermel, 2011; Laursen and Salter, 2006) effects, empirical

studies are reflective of the contradicting theoretical arguments outlined above.

In light of these puzzling findings, this paper attempts to answer the theoretically and practically meaningful question of when internal R&D and external knowledge acquisitions are more or less substitutive (complementary) by examining the role of governance modes as a potentially important contextual variable. More specifically, drawing on the relational view, it can be argued that the substitutive effect of R&D will be smaller under informal governance than formal governance, as the former governance mode is likely to mitigate the NIH syndrome. Indeed, the relational view suggests that the choice of governance arrangements emphasizing either trust or formalization may affect the exchange relationship between collaboration partners quite differently (Dyer and Singh, 1998; Vlaar et al., 2007). In this respect, Das and Teng (1998) propose that the use of formal, contract-based governance mechanisms undermine trust among collaboration partners, whereas informal or social control mechanisms will enhance the level of trust. Contracts are intended to establish control over innovation partners by prescribing their roles and responsibilities, specify penalties for noncompliance, and determine the outcomes to be delivered (Poppo and Zenger, 2002). As such, collaboration partners may perceive contracts as signaling distrust, thereby encouraging rather than discouraging opportunistic behavior (Ghoshal and Moran, 1996; Vlaar et al., 2007).¹ In contrast, informal governance is generally expected to foster trust and cooperative behavior as actors are, for example, likely to perceive the collaboration as more self-determined and reciprocal (e.g., Das and Teng, 1998). Although mutual trust does not constitute a guarantee for trustworthy behavior and coordinated actions among partners (see McEvily, Perrone, and Zaheer, 2003), it is particularly critical for containing the NIH syndrome.² Indeed, mutual trust helps enhance—and is in turn enhanced by—social interactions among employees of partnering firms by such mechanisms as interfirm staff transfers or voluntary learning groups (Dyer and Nobeoka, 2000). Mutual trust and strong personal ties established through interfirm socialization thus hold potential to positively affect employees' willingness not only to share their own knowledge, but also to absorb knowledge inputs from partners (Husted

¹ Scholars also suggest that these negative side effects can potentially be contained by designing effective contracts that may even help establish and maintain mutual trust (Poppo and Zenger, 2002).

² Importantly, these arguments should not be interpreted as pointing to a universal superiority of informal governance. Instead, as proposed in H1 and H2, the configuration of governance modes needs to be tailored to specific characteristics of the partner type and the exchange relationship.

and Michailova, 2002; Van Wijk, Jansen, and Lyles, 2008). Consistent with such arguments, Deci and Ryan's (1985) self-determination theory predicts R&D employees to be less reluctant to use external ideas and technology when such knowledge absorption occurs voluntarily rather than being enforced by hierarchical or contractual mechanisms. For similar reasons, informally governed collaboration is less likely to be perceived as threatening the legitimacy of the R&D department (Grimpe and Kaiser, 2010). Taken together, it can thus be expected that:

H3a: Informally governed, market-focused collaborations will be less susceptible to negative interaction effects with internal R&D than their formally governed counterparts.

H3b: Informally governed, science-focused collaborations will be less susceptible to negative interaction effects with internal R&D than their formally governed counterparts.

Methods

Setting and Data

To test the hypotheses developed above, this study draws on cross-sectoral data from German firms operating in 21 distinct manufacturing and services sectors. Such data were collected as part of the 2008 wave of the Mannheim Innovation Panel (MIP). Of the 18,109 German firms surveyed, 6684 returned usable questionnaires, yielding a response rate of 36.9%. The MIP is an annual survey and constitutes the German contribution to the well-known European Community Innovation Survey (CIS). The CIS is widely considered as a comprehensive source of microlevel data on firms' innovation activities (Laursen and Salter, 2006). Firms' CEO or R&D managers are asked to provide firm-level information on a range of innovation-related themes including innovation expenditures, innovation collaboration, and innovation outcomes. Regularly conducted in all European member states, the CIS adopts a harmonized survey methodology that implements the recommendations of the Oslo Manual (OECD, 1997). Its reliability, validity, and interpretability have been established by means of intensive testing. As a result, CIS data have come to inform a number of insightful studies focusing on innovation collaboration (e.g., Duysters and Lokshin, 2011; Grimpe and Sofka, 2009; Laursen and Salter, 2006; Leiponen and Helfat, 2010; Spithoven, Frantzen, and Clarysse, 2010).

Measures

Dependent variables. Following recent CIS research (Klingebiel and Rammer, 2014), two indicators are employed to measure firms' incremental and radical NPD success. First, the variable *Revenue share from new-to-firm products* captures the estimated percentage share of a firm's total revenues in 2007, which can be attributed to products launched between 2005 and 2007 that are new to the focal firm, but not to its target market. Second, respondents provided estimates of the *Revenue share from new-to-market products*, which captures firms' success in launching and commercializing radically new products that constitute true market novelties. Both measures were subsequently categorized using a 9-point scale ranging from 0 (0%) to 8 (75–100%). Although not without imperfections, such survey-based measures are generally better suited than patent- or R&D-based indicators to capture firms' NPD success rather than their inventive performance. As a result, they enjoy growing acceptance in the field of innovation studies in general (Mairesse and Mohnen, 2002) and the area of open innovation in particular (Klingebiel and Rammer, 2014; Laursen and Salter, 2006; Leiponen and Helfat, 2010).

Independent variables. This study focuses on firms' innovation collaborations with (1) business customers and (2) universities and research institutes as ideal-type representatives of market- and science-focused collaboration, respectively. For each partner type, respondents were asked to indicate not only whether their firm had been engaged in any innovation collaboration between 2005 and 2007, but also whether such collaborations were never, in part, or always contract-based. Based on this information, three governance modes at the heart of the relational view could be distinguished: (1) informal, self-enforcing governance (never contract based); (2) hybrid governance (partly contract based); and (3) formal, contractual governance (always contract based). Six dummy variables were constructed to capture all possible partner type (market- and science-focused) and governance mode (informal, hybrid, and formal) combinations. These variables were labeled *Informal market-focused collaboration*, *Hybrid market-focused collaboration*, *Formal market-focused collaboration*, *Informal science-focused collaboration*, *Hybrid science-focused collaboration*, and *Formal science-focused collaboration*, respectively, and took the value of 1 if the focal firm was engaged in such collaboration and 0 otherwise.

Moderating variable. In line with established practice (e.g., Greve, 2003), firms' internal *R&D intensity* is measured as the ratio of total R&D expenditures to total annual revenues.

Control variables. A number of potentially confounding variables were controlled for. First, differences in *Firm size* measured as the logarithm of the total number of full-time employees were accounted for. Although firm size is expected to influence firms' NPD success, the precise nature of the expected effect remains somewhat ambiguous (Faems et al., 2005). On the one hand, larger firms might demonstrate superior levels of NPD success thanks to economies of scale or risk diversification advantages. On the other hand, small companies tend to be more flexible and adaptive and might thus outperform their larger counterparts when it comes to changing organizational routines and launching new products (Hannan and Freeman, 1984). Second, the percentage of employees holding a university degree measured on an interval scale from 0 (0%) to 8 (> 75%) was employed as a proxy for a firm's *Human capital*. Third, including *Export intensity* provided a means to control for interfirm differences in internationalization and thus in stimulus and demand for new products. Fourth, *Firm location* was controlled for to account for the persistent competitive disadvantage of firms located in East Germany (1) vis-à-vis their counterparts in West Germany (0). Last but not least, a full set of industry dummies was included to capture systematic differences in product innovation success across the 21 distinct groups of two-digit level Nomenclature statistique des activités économiques dans la Communauté européenne (NACE) sectors included in the study (Faems et al., 2005).

Analysis

The two dependent variables *Revenue share from new-to-firm products* and *Revenue share from new-to-market*

products range from 0 (0%) to 8 (75–100%). Both measures are said to be double censored, because they do not allow discriminating among firms located at either the lower end or the upper end of the range captured. Ordinary least squares (OLS) estimates might thus be biased under such circumstances calling for alternative estimation techniques. Following prior CIS-based studies seeking to explain the NPD success of firms in Belgium (Faems et al., 2005), Finland (Leiponen and Helfat, 2010), the United Kingdom (Laursen and Salter, 2006), and Germany (Klingebiel and Rammer, 2014), Tobit maximum likelihood estimation with robust standard errors is employed in all models. This estimator explicitly accounts for the nonlinearity introduced by the double censoring of the dependent variables, thus providing more accurate estimates than OLS. As both measures of NPD success are strongly skewed, the dependent variables were log transformed consistent with prior CIS research (Laursen and Salter, 2006). To allow for a temporal sequencing of independent and dependent variables, a partial lag is introduced by means of the survey design such that information on innovation collaborations refers to the period from 2005 to 2007, while firms' NPD success is captured in 2007.

Results

Results from Descriptive Analyses

After accounting for missing data on the variables included in the model, 2502 German firms remain in the final sample used in all subsequent analyses. Table 1 depicts the diffusion of the three modes of governing market- and science-focused innovation collaborations among these 2502 German firms, of which 1486 (59.4%) were operating in manufacturing sectors and 1016 (40.6%) in service sectors. Table 1 displays the observed frequencies of all six combinations of partner

Table 1. Patterns of Collaborative Innovation

Partner Type	Governance Mode	Full Sample	Subsamples	
			Manufacturing	Services
Market (business customers)	Informal, self-enforcing governance	31.49	33.45	28.64
	Hybrid governance	23.18	26.92	17.72
	Formal, contractual governance	13.43	12.05	15.45
Science (universities and research institutes)	Informal, self-enforcing governance	23.02	22.88	23.23
	Hybrid governance	15.03	17.23	11.81
	Formal, contractual governance	10.23	12.65	6.69

Sample: $N = 2502$ (manufacturing: $N = 1486$; services: $N = 1016$). Figures in percent of firms sampled engaged in specific combination of partner type and governance mode.

Table 2. Descriptive Statistics and Pairwise Correlations

Variable	Mean	Standard Deviation	Minimum	Maximum	1.	2.	3.	4.	5.
1. Revenue share from new-to-firm products	.72	1.55	.00	8.00					
2. Revenue share from new-to-market products	.59	1.47	.00	8.00	.56				
3. Firm size	216.60	1973.94	1.00	75,944.00	.03	.04			
4. R&D intensity	1.54	3.60	.00	15.00	.42	.44	.01		
5. Human capital	3.25	2.44	.00	8.00	.19	.22	.03	.37	
6. Export intensity	.18	.26	.00	.85	.19	.23	.06	.21	.09

Significance: Correlations above .05 are significant at the 5% level. Sample: $N = 2502$. Dummy variables not included.
R&D, research and development.

type (market- and science-focused) and governance mode (informal, hybrid, and formal). Three findings appear particularly noteworthy. First, it becomes apparent that a substantial portion of firms sampled was engaged in market-focused innovation collaborations with business customers (68.1%) or in science-focused innovation collaboration with universities and research institutes (48.3%) between 2005 and 2007. The wide adoption of customer collaborations is little surprising, as prior research has found customers to play a vital role in the NPD process (e.g., von Hippel, 1986). The frequent reliance on universities and research institutes as partners for innovation observed in this sample, however, is remarkable in that it qualifies the misleading impression created by prior research that only a small minority of firms is likely to draw on universities to fuel their innovation activities. Second, descriptive statistics reveal a dominance of informal, self-enforcing forms of governance, which are notably more prevalent than formal, contractual as well as hybrid mechanisms with regard to both market- and science-focused innovation collaborations. Consistent with arguments from the relational view stressing the notable costs and challenges that might be involved in concluding, monitoring, and enforcing contracts, firms thus appear to access valuable need and solution information possessed by customers and universities primarily by means of informal collaborations relying on trust and other self-enforcing mechanisms rather than formal contracts. Third, subsample analyses reveal substantial differences in patterns of collaboration across broad sectors of economic activity. Most notably, manufacturing firms were found to be more likely than service firms to engage in innovation collaborations with both business customers (72.4% compared with 61.8%) and universities or research institutes (52.8% compared with 41.7%). Interestingly, relative to manufacturing firms, service firms relied on formal, contractual governance modes more strongly with regard to market-focused collaborations with business customers, yet less strongly with regard to

science-focused collaborations with universities and research institutes.

Table 2 shows descriptive statistics and pairwise correlations for all continuous variables used in this study. It reveals considerable interfirm variation for all variables including firms' revenue shares from new-to-firm and from new-to-market products, firm size, R&D intensity, human capital, and export intensity. No excessively large correlations among control variables could be detected, pointing to the absence of any notable multicollinearity issues, an assessment supported by a mean variance inflation factor of 2.71 with individual values ranging from a minimum of 1.09 to a maximum of 5.15.

Results from Regression Analyses

Table 3 presents the results from Tobit maximum likelihood regression analyses with robust standard errors explaining differences in firms' log-transformed revenue share from new-to-firm and from new-to-market products. Four models were estimated for each indicator of firms' NPD success to shed light on the complex interplay between partner types, governance modes, and internal R&D. Models 1 and 2 estimate the impact of six combinations of partner type (market- and science-focused) and governance mode (informal, hybrid, and formal) on firms' NPD success. Models 3 and 4 in turn investigate the extent to which the interaction effect between market- and science-focused collaboration on the one hand and internal R&D on the other is sensitive to the governance mode adopted.

Partner type–governance mode configurations and NPD success. Consistent with H1, formal *market-focused* collaborations were found to exhibit a notably stronger positive effect on firms' revenue share from both new-to-firm (Model 1a) and new-to-market products (Model 1b) than informally governed, market-focused collaborations. These differences in coefficient estimates are statistically significant at the 1% level, offering

Table 3. Tobit Analyses Explaining Product Innovation Success

	Model 1		Model 2		Model 3		Model 4	
	Model 1a New-to-Firm Products	Model 1b New-to-Market Products	Model 2a New-to-Firm Products	Model 2b New-to-Market Products	Model 3a New-to-Firm Products	Model 3b New-to-Market Products	Model 4a New-to-Firm Products	Model 4b New-to-Market Products
<i>Control variables</i>								
1 Constant	−4.1371*** (.6217)	−4.6560*** (.5989)	−3.7194*** (.6571)	−4.3413*** (.5582)	−4.1040*** (.6211)	−4.6425*** (.6050)	−3.6039*** (.6630)	−4.2461*** (.5579)
2 Firm size (log)	.1843*** (.0256)	.1911*** (.0304)	.1666*** (.0271)	.1485*** (.0308)	.1818*** (.0259)	.1890*** (.0305)	.1571*** (.0270)	.1380*** (.0307)
3 R&D intensity	.3909*** (.0384)	.4348*** (.0436)	.3891*** (.0402)	.4154*** (.0455)	.6186*** (.0963)	.6618*** (.1072)	.5714*** (.0800)	.5620*** (.0957)
4 Human capital	.0715*** (.0217)	.1189*** (.0268)	.0614*** (.0220)	.1014*** (.0270)	.0721*** (.0217)	.1196*** (.0267)	.0590*** (.0220)	.1004*** (.0269)
5 Export intensity	.3929*** (.1695)	.5105*** (.1924)	.4398*** (.1738)	.5305*** (.1954)	.3754*** (.1698)	.4957*** (.1925)	.4120*** (.1720)	.5075*** (.1946)
6 Firm location	.0571 (.0938)	−.3135*** (.1114)	.0146 (.0951)	−.3676*** (.1130)	.0567 (.0939)	−.3229*** (.1114)	.0119 (.0948)	−.3674*** (.1122)
7 Sector dummy set	Yes***	Yes***	Yes***	Yes***	Yes***	Yes***	Yes***	Yes***
<i>Main effects</i>								
8 Informal market-focused collaboration	.6376*** (.1197)	.6006*** (.1512)	.6233*** (.1220)	.6160*** (.1550)	.6233*** (.1220)	.6160*** (.1550)	.2568*** (.1108)	.3620*** (.1337)
9 Hybrid market-focused collaboration	1.1914*** (.1177)	1.3248*** (.1439)	1.2188*** (.1201)	1.3420*** (.1491)	1.2188*** (.1201)	1.3420*** (.1491)	.7673*** (.1187)	1.0839*** (.1378)
10 Formal market-focused collaboration	1.0538*** (.1417)	1.2844*** (.1654)	1.0987*** (.1489)	1.3542*** (.1731)	1.0987*** (.1489)	1.3542*** (.1731)	.9100*** (.1418)	1.2641*** (.1635)
11 Informal science-focused collaboration			.2889*** (.1092)	.4029*** (.1316)				
12 Hybrid science-focused collaboration			.6807*** (.1172)	1.0294*** (.1361)				
13 Formal science-focused collaboration			.7644*** (.1327)	1.0574*** (.1553)				
<i>Interaction effects</i>								
14 Informal market-focused collaboration × R&D intensity			−.1329 (.1174)		−.1329 (.1174)	−.2491* (.1358)		
15 Hybrid market-focused collaboration × R&D intensity			−.2892*** (.1073)		−.2892*** (.1073)	−.2307* (.1180)		
16 Formal market-focused collaboration × R&D intensity			−.2810** (.1112)		−.2810** (.1112)	−.2825** (.1246)		
17 Informal science-focused collaboration × R&D intensity							.0903 (.1161)	.1373 (.1312)
18 Hybrid science-focused collaboration × R&D intensity							−.2943*** (.0975)	−.2074* (.1132)
19 Formal science-focused collaboration × R&D intensity							−.3104*** (.0983)	−.3173*** (.1170)
Total observations	2502	2502	2502	2502	2502	2502	2502	2502
Uncensored observations	607	465	607	465	607	465	607	465
Log likelihood	−1729.66	−1414.12	−1762.77	−1432.13	−1725.03	−1411.63	−1751.50	−1423.303
F-test statistic	33.7674***	33.6595***	33.8703***	35.0302***	30.7293***	30.2782***	32.3800***	33.7147***
McFadden Pseudo R ²	.1678	.1987	.1518	.1885	.1700	.2001	.1572	.1935

* $p < .10$; ** $p < .05$; *** $p < .01$.

Double-censored Tobit estimator employed in all models. Marginal effects calculated at the means of all the independent variables reported. Robust standard errors reported in parentheses. Continuous variables used to calculate the interaction terms were standardized. R&D, research and development.

Table 4. Overview of Results

Hypothesis		Support
H1	<i>Formally</i> governed, <i>market-focused</i> collaborations will be associated more strongly with firms' incremental and radical NPD success than their <i>informally</i> governed counterparts.	√
H2	<i>Informally</i> governed, <i>science-focused</i> collaborations will be associated more strongly with firms' incremental and radical NPD success than their formally governed counterparts.	x
H3a	<i>Informally</i> governed, <i>market-focused</i> collaborations will be less susceptible to negative interaction effects with internal R&D than their formally governed counterparts.	(√)
H3b	<i>Informally</i> governed, <i>science-focused</i> collaborations will be less susceptible to negative interaction effects with internal R&D than their formally governed counterparts.	√

NPD, new product development; R&D, research and development.

support for H1. H2 suggested that informal rather than formal modes of governance would be more conducive to translating science-focused collaborations into higher NPD success. Contrary to theoretical expectations, however, coefficient estimates for formally governed science-focused collaborations significantly exceed those of their informally governed counterparts with regard to firms' revenue share from both new-to-firm (Model 2a) and new-to-market products (Model 2b). H2 is thus not supported.

Open innovation, internal R&D, and collaboration governance. H3a and H3b pertained to the interaction effect between firms' internal R&D intensity and their engagement in market- and science-focused innovation collaborations. In particular, it was suggested that informal governance relying on trust and other self-enforcing mechanisms might make the focal firm less vulnerable to possible negative side effects associated with the simultaneous pursuit of internal R&D and external collaboration. To test both hypotheses, interaction terms between internal R&D and all six combinations between partner type and governance mode were computed. The coefficient estimates for these interaction terms in Models 3 and 4 were expected to be less substitutive in character for informally governed collaborations than for formally governed collaborations. As for firms' revenue share from new-to-firm products, the interaction effect between *formal* market-focused collaboration and R&D intensity is negative and statistically significant (Model 3a: $\beta = -.281$, $p < .01$), whereas the interaction effect between *informal* collaboration and R&D intensity is notably higher and statistically insignificant (Model 3a: $\beta = -.133$, not significant [n.s.]). With regard to firms' revenue share from new-to-market products, however, the coefficient estimates for informal and formal market-focused collaboration are both statistically significant with their difference failing to achieve statistical significance. H3a is thus partly supported.

Similarly, internal R&D exhibits a highly significant negative interaction effect with *formal* science-focused collaboration with regard to both indicators of firms' NPD success (Model 4a: $\beta = -.310$, $p < .01$; Model 4b: $\beta = -.317$, $p < .01$). The interaction effects with regard to *informal* science-focused collaboration in contrast are positive and statistically insignificant in both models (Model 4a: $\beta = .090$, n.s.; Model 4b: $\beta = .137$, n.s.). H3b is therefore supported. Table 4 provides a concise summary of the main findings.³

Results from Post Hoc Analyses

To examine the robustness of the findings presented above, a number of additional analyses were performed. First, all models were reestimated using standard OLS and ordered probit estimators instead of a Tobit maximum likelihood estimator, yielding a consistent pattern of results for all effects of interest including the interaction effects. Second, a split-sample approach was employed as an alternative means to examine the interaction effects between internal R&D and external collaboration. This reanalysis is potentially valuable, as the true interaction effect is not always appropriately described by the value of the Tobit coefficient estimates for the interaction term (Wiersema and Bowen, 2009). Most notably, it revealed that high R&D spenders benefit significantly more strongly than low R&D spenders from science-focused collaborations provided these are informally rather than formally governed—a finding that

³ As shown in Table 3, the hybrid governance mode reflecting the flexible use of both formal and informal governance mechanisms by the focal firm was found to be significantly and positively related to its NPD success for all possible partner type–innovation objectives combinations, while exhibiting effect sizes that exceed those of informal governance and are comparable with those of formal governance (Models 1a, 1b, 2a, and 2b). A similar pattern can be observed for the interaction effects with internal R&D, i.e., there is a substitutive relationship between innovation collaboration based on hybrid governance and the overall effect sizes reported for hybrid governance being slightly lower than those reported for purely formal governance in Models 3b, 4a, and 4b.

underlines the robustness of the results reported above. Third, the extent to which the results are robust to the omission of specific control variables was examined by sequentially eliminating individual control variables—a process that led to consistent results. Fourth, additional analyses were conducted to explore suppliers as a third partner type distinguished in the literature (Sofka and Grimpe, 2010). Results for suppliers of materials and services (available upon request) are largely consistent with those reported for business customers. Fifth, the relationship between the breadth of a firm's overall collaboration portfolio and its reliance on contractual formalization of collaborative relationships was examined. Results reveal a substantial positive association between these constructs suggesting that formal, contract-based governance mechanisms gain in salience both in absolute terms and relative to informal, self-enforcing mechanisms as the collaboration portfolio grows larger becoming increasingly difficult to govern informally. Finally, possible interindustry differences in the observed effects were explored by comparing model estimates for manufacturing and service subsamples. Service firms appear less inclined than manufacturing firms to establish informal and formal collaborative agreements with market actors and in particular with the science base. This stands in stark contrast to the fact that the beneficial effects in particular of formally governed market- and science-focused collaboration were even more pronounced among service than manufacturing firms. Service firms might therefore be well advised to foster their collaborative ties with outside actors to close the collaboration gap vis-à-vis their manufacturing counterparts. Moreover, it is worth noting that the substitutive effects between market-focused collaboration and internal R&D observed in the full sample are notably less pronounced among service firms, for which all interaction terms remain statistically insignificant.

Discussion

Given mounting evidence on the performance-enhancing effects of innovation collaboration, opening up the innovation process is increasingly perceived as a vital requirement instead of an optional luxury (Lichtenthaler, 2011). Recent research, however, has sought to temper the growing enthusiasm finding that returns to openness are not only declining at the margin, but also dependent on the presence of certain boundary conditions both at the project level (Salge et al., 2013) and at the firm-level (Laursen and Salter, 2006; Salge et al., 2012). This study adds to this emerging contingency literature on open

innovation (Huizingh, 2011). It does so by untangling how the complex interplay among partner types, governance modes, and internal R&D affects the attainment of firms' innovation objectives. Findings based on cross-industry data from 2502 German firms revealed that the positive payoff from innovation collaboration with business customers and universities varies as a function of the specific governance mode employed. Formal, contractual governance mechanisms turned out to be more effective than informal, self-enforcing mechanisms in enhancing firms' returns not only from market-focused collaboration, but remarkably also from science-focused collaboration. Informal, relational governance, in contrast, emerged as being more effective than formal, contractual governance in safeguarding the focal firm from possible negative interaction effects between external collaboration and internal R&D. The conceptual arguments and empirical findings on the twofold moderating role of collaboration governance presented in this study have a number of implications for innovation research and practice.

Implications for Research

This study advances the rapidly growing literature on open innovation in at least three important ways. First, this study introduces the relational view (Dyer and Singh, 1998) as a theoretical framework to better ground open innovation research conceptually. This is a meaningful contribution, as the field is still in need of a strong theoretical foundation with the potential to integrate a broad range of conceptual ideas stemming from interorganizational knowledge sharing and related literatures (Lichtenthaler, 2011). The relational view is particularly well suited to fill the current theoretical void, as it emphasizes the role of complementary resources, relationship-specific assets, knowledge-sharing routines, and governance mechanisms (Dyer and Singh, 1998; Lavie, 2006), all of which are pivotal to open innovation research. Given its reliance on ideas from resource-based thinking and transaction cost economics, the relational view provides a valuable conceptual toolkit for the analysis of interorganizational exchange relationships. It thus holds potential to inform research not only on collaboration governance as illustrated in this study, but also on a broad range of related topics of potential relevance for the emerging contingency literature on open innovation. Promising candidates include for instance the moderating role of prior partner-specific experience (Love et al., 2013), relationship-specific investments (Kang, Mahoney, and Tan, 2009), incentives and routines for

interfirm knowledge sharing (Dyer and Nobeoka, 2000; Fu, 2012), or interfirm cultural and procedural compatibility (Emden et al., 2006).

Second, this study introduces collaboration governance as a critical, yet underexplored contingency factor affecting the payoff from open innovation. Most notably, this study revealed that formal and informal governance exhibit complementary benefits (Poppo and Zenger, 2002). While formal governance emerged as being more effective at enhancing a firm's direct payoff from innovation collaboration, informal governance turned out to better safeguard the focal firm from possible substitution effects between external collaboration and internal R&D. Although at odds with H2 and the widespread belief in the general superiority of informal governance propagated by the relational view, this finding is not unexpected (Poppo, Zhou, and Zenger, 2008). Dyer and Singh (1998) themselves identified informal, self-enforcing mechanisms such as relational trust as being highly reliant on time and repeated interactions to materialize. Formal contracts, in contrast, allow for a standardization of policies and procedures governing exchange relations across a range of structurally similar collaborations, thus economizing on time and time-related transaction costs (Williamson, 1985).⁴ Given the substantial breadth and continuous reconfiguration of many firms' collaboration portfolios (Faems et al., 2005; Srivastava and Gnyawali, 2011), this might limit the use and relative effectiveness of informal safeguards in enhancing the payoff from collaboration not only with market-focused partners, but also with science-focused partners.⁵ In this respect, it should be noted that the relative advantage of formal governance observed in this study might have been magnified by the specific German institutional context, which is characterized by a strong belief in the efficacy and legal enforceability of contracts (Arrighetti, Bachmann, and Deakin, 1997). Overall, the findings reported in this study not only contribute to the open innovation literature, but

also inform extant relational view theorizing. In particular, new insights into the ongoing debate of whether trust and formalization act as substitutes (e.g., Dyer and Singh, 1998) or complements (e.g., Poppo and Zenger, 2002) are provided.

Third, this study contributes to solving the puzzle on the interplay between external collaboration and internal R&D that continues to engage product innovation scholars (e.g., Hagedoorn and Wang, 2012; Vega-Jurado et al., 2009). Confronted with mixed empirical evidence and countervailing theoretical arguments informed by either absorptive capacity or NIH research, this study sought to reconcile both positions, i.e., those uncovering complementarities (Cassiman and Veugelers, 2006; Grimpe and Kaiser, 2010; Salge et al., 2012) and those unearthing substitution effects (Caloghirou, Kastelli, and Tsakanikas, 2004; Laursen and Salter, 2006) among both activities. More specifically, the findings presented in this paper suggest that informal collaboration governance relying on relational trust, partnership reputation, and interfirm socialization safeguards the focal firm from possible substitution effects and enables them to harness the value from both internal R&D and external collaboration. A reliance on formal, contract-based governance, in contrast, appears to render the simultaneous pursuit of internal R&D and external collaboration more challenging. This finding can be interpreted as underlining the importance of relational trust as a critical means to foster employees' willingness to absorb partner knowledge inputs thereby containing, if not overcoming, the NIH syndrome. It is thus not only the level of internal R&D (Hagedoorn and Wang, 2012), but also the specific governance mechanism employed that determines the relative importance of complementarity and substitution effects between internal R&D and external collaboration. This study thus addresses Cassiman and Veugelers's (2006) call to shift the debate toward identifying the specific conditions under which complementarity or substitution effects tend to prevail, instead of arguing for the general dominance of one or the other. This more subtle approach is not only more appropriate conceptually, but also expected to be more meaningful for product innovation managers seeking practical advice on how to optimize their concurrent engagement in internal R&D and external collaboration.

Implications for Practice

This study offers several meaningful insights for decision-makers in innovation management and policy. At a more general level, this study adds to the now

⁴ The coordination function of contracts is one potential explanation as to why informal governance was not found to be the more-effective governance mode for science-based collaboration, as proposed in H2. By decomposing tasks and clarifying the objectives, roles, and responsibilities in a relationship, contracts can help to overcome coordination problems arising from the cognitive limitations of individuals (Carson et al., 2006; Vlaar et al., 2007). As universities and research institutes differ substantially from business firms in their structures, processes, cultures, and interpretive schemes of employees, a coordination of activities with these partners might be particularly challenging for the focal firm.

⁵ Additional findings reveal a strong NPD-enhancing effect of the hybrid governance mode, which can be interpreted as a firm's customized use of both contract-based and relational governance mechanisms depending on the particular collaboration at hand. This underlines the importance of going beyond pure governance archetypes to examine more-complex configurations—a task that is beyond the scope of this study and remains to be addressed by future research.

substantial evidence base on the performance-enhancing effects of open innovation (West and Bogers, 2014), thereby underlining its strategic relevance. Importantly, however, the findings presented in this paper indicate that positive returns to innovation collaboration do not accrue automatically. Instead, innovation managers and policy-makers need to play an active role in facilitating the identification, assimilation, and utilization of external knowledge, if they are to help organizations reap the benefits of open innovation (Salge et al., 2013). While prior firm-level research has highlighted the importance of partner-type selection (Sofka and Grimpe, 2010), innovation management (Salge et al., 2012), and internal constraints (Garriga et al., 2013) in affecting firms' open innovation payoff, this study alerts decision-makers to the critical role of collaboration governance (Poppo and Zenger, 2002). Innovation managers thus face the challenging task of tailoring governance mechanisms not only to their innovation objectives, but also to the characteristics of the specific partner type and exchange relationship. Policy-makers in turn are charged with fostering the innovation management capabilities within public sector organizations in fields such as education, health-care, or central government, which become an increasingly important part of firms' portfolio of innovation collaborations (Lee, Hwang, and Choi, 2012). Moreover, policy-makers across the world need to maintain—and if necessary (re)establish—the effectiveness of formal governance mechanisms, which hinges upon adequate legal institutions to ensure the legitimacy, adaptability, and enforceability of contracts (Zhou and Poppo, 2010).

At a more specific level, this study not only sensitizes decision-makers to possible substitution effects between internal R&D and external collaboration, but also provides them with actionable knowledge to contain the underlying NIH syndrome (Katz and Allen, 1982). In particular, this study suggests that informal governance mechanisms such as relational trust, interfirm socialization, and partnership reputation are likely to positively affect employees' attitudes toward external knowledge, thus reducing the tendency to reject such inputs as generally inferior. Whenever the threat of internal resistance toward external knowledge risks compromises a firm's open innovation efforts, innovation managers might be well advised to rely more strongly on informal governance. This might be the case especially in early stages of the transition from a closed to an open innovation model, when the internal readiness for open innovation is still being developed (Chiaroni, Chiesa, and Frattini, 2011). In later stages of a firm's open innovation journey, in contrast, the relative time and transaction cost advantages of

formal governance might materialize more fully, making carefully crafted collaboration contracts the governance mode of choice.

Limitations and Future Research

Notwithstanding its theoretical and managerial contributions, this study is not without its own limitations, which might serve as a starting point for future research. First and perhaps most importantly, this paper dichotomizes prominent dimensions characterizing firms' innovation collaborations. This holds true for *partner types* (market and science focused), *governance modes* (informal, self-enforcing, and formal; contractual collaboration) and *innovation objectives* (incremental and radical). Although grounded in extant literature and useful for analytical purposes, this approach captures only archetypical instances of each dimension and falls short of doing full justice to the multifaceted nature of collaborative arrangements firms forge to enhance their NPD success. It thus remains an important task for future research to include additional partner types such as lead users (von Hippel, 1986), competitors (Gnyawali and Park, 2011), or consultancies (Tether and Tajar, 2008) as well as additional innovation objectives such as successful process innovation (Robertson, Casali, and Jacobson, 2012) or business model innovation (Chesbrough, 2010). In particular, more granular data and analyses are needed to go beyond the pure governance archetypes that feature prominently in the relational view and explore the manifold and often subtle configurations of formal and informal elements of collaboration governance. Such efforts appear especially valuable in light of growing evidence suggesting that formal and informal governance might act as complements rather than substitutes (Poppo and Zenger, 2002). In the best of cases, informal and formal governance will be mutually reinforcing, such that well-specified contracts facilitate the development of trust among partners, with trust in turn acting both as a safeguard against hazards poorly protected by incomplete contracts and a means to ease the adjustment of contracts over time (Poppo and Zenger, 2002). Exploring the extent to which these more-complex configurations of formal and informal mechanisms truly unite the advantages from both governance modes with regard to firms' NPD success thus clearly remains an important task for subsequent contingency studies on open innovation.

Second, future research applying longitudinal or qualitative research designs is needed to complement this study and help overcome some of its methodological

limitations. Panel data research covering longer time periods, for instance, holds the potential to unearth the temporal dynamics that are likely to shape the interplay among partner types, governance modes, and innovation objectives over time. Such research is likely to provide valuable insights into the role of relationship-specific experience—a factor that figures prominently in the relational view (Dyer and Singh, 1998) and was found to give rise to learning effects as part of which firms gradually tailor their portfolio of collaborative linkages and corresponding governance modes to their specific needs (Love et al., 2013). Similarly, in-depth qualitative research—ideally adopting an ethnographic approach (e.g., De Rond and Bouchikhi, 2004)—could complement this study by unpacking the internal workings and social complexities likely to characterize the process of configuring and adapting governance mechanisms over time.

Third, this study is restricted in its conceptual and empirical scope. Conceptually, it examines the interplay between partner-type and collaboration mode as two particularly salient dimensions that characterize firms' collaborative arrangements. That said, further attributes are worthy of dedicated study especially when examined as interacting with a range of governance modes. Promising candidates include *collaboration purpose* and *collaboration timing*, which would allow for a distinction between exploitative and explorative collaborations (Rothaermel and Deeds, 2004) as well as between early-stage and late-stage collaborations (Olson, Walker, Ruekerf, and Bonnerd, 2001), respectively. Empirically, this study remains constrained to the specific economic and institutional context of Germany. Decision-makers in Germany were found to exhibit notably higher confidence in the efficiency and stability of the legal system, greater knowledge of contract law, and a stronger reliance on long-term interfirm contracts than their counterparts in other countries such as Italy or Great Britain (Arrighetti et al., 1997). This is potentially consequential for the relationships of interest, as stronger beliefs in the legal enforceability of contracts tend to be associated with a greater use and arguably also greater effectiveness of formal vis-à-vis informal governance (Zhou and Poppo, 2010). Comparative, cross-country research is thus needed to explore the extent to which the findings especially on the role of collaboration governance presented in this study vary across legal contexts.

There are hence several promising opportunities for future research to deepen scholarly understanding of the interplay among external partner types, interfirm governance modes, and internal R&D, thereby further fueling the emerging contingency literature on open innovation.

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